

3.9 Exponential & Logarithmic Derivatives Worksheet

1. Given $g(t) = e^{t^3}$, find its second derivative $g''(t)$.

$$g'(t) = e^{t^3} \cdot 3t^2$$

$$g''(t) = 3t^2 e^{t^3} (3t^2) + e^{t^3} (6t) = 9t^4 e^{t^3} + 6te^{t^3} = 3te^{t^3} (3t^3 + 2)$$

2. Compute $f'(r)$ given that $f(r) = \tan^{-1}(\sqrt{r^2 + 5})$.

$$f'(r) = \frac{1}{(\sqrt{r^2 + 5})^2 + 1} \cdot \frac{1}{2}(r^2 + 5)^{-1/2}(2r) = \frac{r}{(r^2 + 6)\sqrt{r^2 + 5}}$$

3. Given $h(t) = \ln(t^3 + 4t^2 + 6)$, find its second derivative $h''(t)$.

$$h'(t) = \frac{1}{t^3 + 4t^2 + 6} (3t^2 + 8t) = \frac{3t^2 + 8t}{t^3 + 4t^2 + 6}$$

$$h''(t) = \frac{(t^3 + 4t^2 + 6)(6t + 8) - (3t^2 + 8t)(3t^2 + 8t)}{(t^3 + 4t^2 + 6)^2}$$

4. Compute $h'(t)$ if $h(t) = \ln(\tan(t^5 + 2t))$.

$$h'(t) = \frac{1}{\tan(t^5 + 2t)} \sec^2(t^5 + 2t)(5t^4 + 2)$$

5. Compute $f'(x)$ if $f(x) = 4^x \cdot \sin(6x)$.

$$f'(x) = 4^x \ln(4) \sin(6x) + 4^x \cos(6x) 6 = 4^x [\ln(4) \sin(6x) + 6 \cos(6x)]$$

6. Compute $\frac{dy}{dx}$ given each of the following equations.

a) $y = \sqrt{\arcsin(e^{-5x})}$

$$\frac{dy}{dx} = \frac{1}{2} [\arcsin(e^{-5x})]^{-1/2} \cdot \frac{1}{\sqrt{1-(e^{-5x})^2}} \cdot e^{-5x} \cdot (-5)$$

b) $y = (x^2 + 9)^{3\ln(x)}$

We will first rewrite the given equation: $\ln(y) = \ln[(x^2 + 9)^{3\ln(x)}] \rightarrow \ln(y) = 3\ln(x) \cdot \ln[x^2 + 9]$

Now we take the derivative implicitly:

$$\begin{aligned} \frac{d}{dx}[\ln(y)] &= \frac{d}{dx}[3\ln(x) \cdot \ln(x^2 + 9)] \rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{3}{x} \ln(x^2 + 9) + 3\ln(x) \frac{1}{x^2 + 9} (2x) \\ &\rightarrow \frac{dy}{dx} = y \left[\frac{3\ln(x^2 + 9)}{x} + \frac{6x\ln(x)}{x^2 + 9} \right] \\ &\rightarrow \frac{dy}{dx} = (x^2 + 9)^{3\ln(x)} \left[\frac{3\ln(x^2 + 9)}{x} + \frac{6x\ln(x)}{x^2 + 9} \right] \end{aligned}$$

c) $f(x) = \ln(\sqrt[3]{\sin(x^5 + 4x)})$

$$f'(x) = \frac{1}{\sqrt[3]{\sin(x^5 + 4x)}} \cdot \frac{1}{3} [\sin(x^5 + 4x)]^{-2/3} \cdot \cos(x^5 + 4x) \cdot (5x^4 + 4)$$

d)

e) $y = \frac{(3x^2 - 1)^2}{\sin(e^x)}$

We can rewrite the function as $y = (3x^2 - 1)^2 \csc(e^x)$.

Therefore, the derivative is calculated as:

$$\begin{aligned} y' &= 2(3x^2 - 1)(6x) \csc(e^x) + (3x^2 - 1)^2 [-\csc(e^x) \cot(e^x)] e^x \\ &= \csc(e^x) (3x^2 - 1) [12x - e^x (3x^2 - 1) \cot(e^x)] \end{aligned}$$

f) $g(t) = \sin^{-1}(e+2t)$

$$g'(t) = \frac{1}{\sqrt{1-(e+2t)^2}}(2) = \frac{2}{\sqrt{1-(e+2t)^2}}$$

g) $h(x) = \log_4(7^{\pi x})$

$$h'(x) = \frac{1}{7^{\pi x} \ln(4)} \cdot 7^{\pi x} \ln(7) \cdot \pi = \frac{\pi \ln(7)}{\ln(4)}$$

h) $y = [\sin(x)]^{5x^2}$

We will first rewrite the given equation: $\ln(y) = \ln\left[(\sin(x))^{5x^2}\right] \rightarrow \ln(y) = 5x^2 \ln(\sin(x))$

Now we take the derivative implicitly:

$$\begin{aligned} \frac{d}{dx}[\ln(y)] &= \frac{d}{dx}[5x^2 \ln(\sin(x))] \rightarrow \frac{1}{y} \frac{dy}{dx} = 10x \ln(\sin(x)) + 5x^2 \frac{1}{\sin(x)} \cos(x) \\ &\rightarrow \frac{dy}{dx} = y[10x \ln(\sin(x)) + 5x^2 \cot(x)] \\ &\rightarrow \frac{dy}{dx} = [\sin(x)]^{5x^2} [10x \ln(\sin(x)) + 5x^2 \cot(x)] \end{aligned}$$

i) $y = [\ln(x)]^{\cos(x)}$

We apply logarithmic differentiation.

$y = [\ln(x)]^{\cos(x)} \Rightarrow \ln(y) = \ln\left([\ln(x)]^{\cos(x)}\right) \Rightarrow \ln(y) = \cos(x) \ln(\ln(x))$. Then, we have that:

$$\begin{aligned} \ln(y) &= \cos(x) \ln(\ln(x)) \Rightarrow \frac{d}{dx}[\ln(y)] = \frac{d}{dx}[\cos(x) \ln(\ln(x))] \\ &\Rightarrow \frac{1}{y} \frac{dy}{dx} = -\sin(x) \ln(\ln(x)) + \cos(x) \frac{1}{\ln(x)} \cdot \frac{1}{x} \\ &\Rightarrow \frac{dy}{dx} = y \left[-\sin(x) \ln(\ln(x)) + \frac{\cos(x)}{x \ln(x)} \right] \\ &\Rightarrow \frac{dy}{dx} = [\ln(x)]^{\cos(x)} \left[\frac{\cos(x)}{x \ln(x)} - \sin(x) \ln(\ln(x)) \right]. \end{aligned}$$

j) $\ln(y^3) = x^3 + y^5$

We take this derivative implicitly:

$$\begin{aligned}\frac{d}{dx}[\ln(y^3)] &= \frac{d}{dx}[x^3 + y^5] \rightarrow \frac{1}{y^3} \cdot 3y^2 \cdot \frac{dy}{dx} = 3x^2 + 5y^4 \frac{dy}{dx} \\ &\rightarrow \frac{3}{y} \frac{dy}{dx} - 5y^4 \frac{dy}{dx} = 3x^2 \\ &\rightarrow \frac{dy}{dx} \left(\frac{3}{y} - 5y^4 \right) = 3x^2 \\ &\rightarrow \frac{dy}{dx} = \frac{3x^2}{(3y^{-1} - 5y^4)} \quad \text{or} \quad \frac{3x^2 y}{(3 - 5y^5)}\end{aligned}$$

7. Use logarithmic differentiation to find the derivative of $y = \frac{x^3 \sqrt{x^4 + 3}}{(5x+1)^8}$.

We rewrite the function as:

$$\begin{aligned}\ln(y) &= \ln\left(\frac{x^3 \sqrt{x^4 + 3}}{(5x+1)^8}\right) \rightarrow \ln(y) = \ln(x^3 \sqrt{x^4 + 3}) - \ln((5x+1)^8) \\ &\rightarrow \ln(y) = \ln(x^3) + \ln(\sqrt{x^4 + 3}) - \ln((5x+1)^8) \\ &\rightarrow \ln(y) = 3\ln(x) + \frac{1}{2}\ln(x^4 + 3) - 8\ln(5x+1)\end{aligned}$$

We now take the derivative:

$$\begin{aligned}\frac{d}{dx}[\ln(y)] &= \frac{d}{dx}\left[3\ln(x) + \frac{1}{2}\ln(x^4 + 3) - 8\ln(5x+1)\right] \\ \frac{1}{y} \frac{dy}{dx} &= \frac{3}{x} + \frac{1}{2} \cdot \frac{1}{x^4 + 3} (4x^3) - \frac{8}{5x+1} (5) \\ \frac{dy}{dx} &= y \left[\frac{3}{x} + \frac{2x^3}{x^4 + 3} - \frac{40}{5x+1} \right] \\ \frac{dy}{dx} &= \frac{x^3 \sqrt{x^4 + 3}}{(5x+1)^8} \left[\frac{3}{x} + \frac{2x^3}{x^4 + 3} - \frac{40}{5x+1} \right]\end{aligned}$$

8. Find $g'(x)$ if $g(x) = \cos^{-1}(e^{2x})$.

$$g'(x) = \frac{-1}{\sqrt{1 - (e^{2x})^2}} (e^{2x})(2) = \frac{-2e^{2x}}{\sqrt{1 - e^{4x}}}$$